

Sequential File Allocation

```
#include <stdio.h>
#include <stdlib.h>
int main() {
    int disk[50] = {0};
    int startBlock, len, i, j, flag;
    char choice;
    printf("--- Sequential File Allocation ---\n");
    do {
        flag = 0;
        printf("\nEnter starting block (0-49): ");
        scanf("%d", &startBlock);
        printf("Enter length of file: ");
        scanf("%d", &len);
        if (startBlock + len > 50) {
            printf("Error: File exceeds disk space.\n");
        } else {
            for (i = startBlock; i < startBlock + len; i++) {
                if (disk[i] == 1) {
                    flag = 1; // Block already allocated
                    break;
                }
            }
            if (flag == 1) {
                printf("Error: Contiguous blocks are not free.\n");
            } else {
                for (j = startBlock; j < startBlock + len; j++) {
                    disk[j] = 1;
                }
                printf("File allocated successfully from block %d to %d.\n", startBlock, startBlock + len - 1);
            }
        }
        printf("Do you want to allocate another file? (y/n): ");
        scanf(" %c", &choice);
    } while (choice == 'y' || choice == 'Y');
    return 0;
}
```

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o pratyushojha@Pratyushs-MacBook-Pro Coding % cd "/Users/pratyushojha/Documents/Coding/" && gcc 8a.c -o 8a && "/U
sers/pratyushojha/Documents/Coding/"8a
1WA24CS216--- Sequential File Allocation ---

Enter starting block (0-49): 4
Enter length of file: 6
File allocated successfully from block 4 to 9.
Do you want to allocate another file? (y/n): Y

Enter starting block (0-49): 5
Enter length of file: 3
Error: Contiguous blocks are not free.
Do you want to allocate another file? (y/n): 

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Indexed File Allocation

```

#include <stdio.h>
int main() {
int disk[50] = {0};
int indexBlock, n, i, flag;
int blocks[50];
char choice;
printf("--- Indexed File Allocation ---\n");
do {
printf("\nEnter index block (0-49): ");
scanf("%d", &indexBlock);
if (disk[indexBlock] == 1) {
printf("Error: Index block %d is already allocated.\n", indexBlock);
} else {
printf("Enter number of blocks for the file: ");
scanf("%d", &n);
printf("Enter the %d block numbers: ", n);
flag = 0; for (i = 0; i < n; i++) {
scanf("%d", &blocks[i]);
if (disk[blocks[i]] == 1) {
flag = 1;
}
}
if (flag == 1) {
} else {
disk[indexBlock] = 1;
for (i = 0; i < n; i++) {
disk[blocks[i]] = 1;
printf("Error: One or more blocks are already allocated. Allocation failed.\n");

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}
printf("File allocated successfully.\n");
printf("Index Block: %d\n", indexBlock);
printf("Allocated Data Blocks: ");
for (i = 0; i < n; i++) {
printf("%d ", blocks[i]);
}
printf("\n");
}
}
printf("Do you want to allocate another file? (y/n): ");
scanf(" %c", &choice);
}
while (choice == 'y' || choice == 'Y');
return 0;

```

```

○ pratyushojha@Pratyushs-MacBook-Pro Coding % cd "/Users/pratyushojha/Documents/Coding/" && gcc 8b.c -o 8b && "/U
sers/pratyushojha/Documents/Coding/"8b
1WA24CS216--- Indexed File Allocation ---

Enter index block (0-49): 8
Enter number of blocks for the file: 4
Enter the 4 block numbers: 12 3 45 31
File allocated successfully.
Index Block: 8
Allocated Data Blocks: 12 3 45 31
Do you want to allocate another file? (y/n): y

Enter index block (0-49): 12
Error: Index block 12 is already allocated.
Do you want to allocate another file? (y/n): █

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}

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Linked File Allocation

```

#include <stdio.h>
#include <stdlib.h>
struct Block {

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int allocated;
int next;
};
int main() {
    struct Block disk[50];
    int startBlock, n, i, currentBlock, nextBlock;
    char choice;
    for (i = 0; i < 50; i++) {
        disk[i].allocated = 0;
        disk[i].next = -1;
    }
    printf("--- Linked File Allocation ---\n");
    do {
        printf("\nEnter starting block (0-49): ");
        scanf("%d", &startBlock);
        if (disk[startBlock].allocated == 1) {printf("Error: Starting block is already allocated.\n");
        } else {
            printf("Enter number of blocks required: ");
            scanf("%d", &n);
            disk[startBlock].allocated = 1;
            currentBlock = startBlock;
            if(n == 1) {
                printf("File allocated successfully. Chain: %d -> NULL\n", startBlock);
            }
            else {
                printf("Enter the remaining %d blocks to form the chain:\n", n - 1);
                int success = 1;
                for (i = 1; i < n; i++) {
                    printf("Next block after %d: ", currentBlock);
                    scanf("%d", &nextBlock);
                    if (disk[nextBlock].allocated == 1) {
                        printf("Error: Block %d is already allocated! Aborting chain.\n", nextBlock);
                        success = 0;
                        break;
                    } else {
                        disk[nextBlock].allocated = 1;
                        disk[currentBlock].next = nextBlock;
                        currentBlock = nextBlock;
                    }
                }
            }
        }
    }
}

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if (success) {
printf("File allocated successfully. Chain: ");
int temp = startBlock;
while (temp != -1) {
printf("%d -> ", temp);
temp = disk[temp].next;
}
printf("NULL\n");
}
}
}
printf("Do you want to allocate another file? (y/n): ");
scanf(" %c", &choice);
}while (choice == 'y' || choice == 'Y');
return 0;
}

```

```

○ pratyushojha@Pratyushs-MacBook-Pro Coding % cd "/Users/pratyushojha/Documents/Coding/" && gcc 8c.c -o 8c && "/U
sers/pratyushojha/Documents/Coding/"8c
1WA24CS216
--- Linked File Allocation ---

Enter starting block (0-49): 3
Enter number of blocks required: 4
Enter the remaining 3 blocks to form the chain:
Next block after 3: 15
Next block after 15: 37
Next block after 37: 1
File allocated successfully. Chain: 3 -> 15 -> 37 -> 1 -> NULL
Do you want to allocate another file? (y/n): █

```